

Tom George

Post-doctoral fellow in ML and theoretical neuroscience,
also building open-source software for science

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ACADEMIC POSITIONS	Post-doctoral Fellow and Canadian Neuroanalytic Scholar, Mila Research on foundation models for neuroscience and NeuroAI with Prof. Blake Richards (McGill/Mila) and Prof. Guillaume Lajoie (UdeM/Mila).	2025–
EDUCATION	PhD, Sainsbury Wellcome Centre, UCL Advisors: Prof. Caswell Barry (UCL), Prof. Claudia Clopath (Imperial) and Dr. Kimberly Stachenfeld (Google DeepMind/Columbia). Member of Google DeepMind NeuroLab . Herchel Smith Scholarship, Harvard University Prestigious and fully funded scholarship to study at Harvard; 4.0 GPA. Advisors: Prof. Cengiz Pehlevan & Prof. Sam Gershman. BA, MA & MSci, Physics (Natural Sciences), University of Cambridge 1 st Class (Part III), 1 st Class (Part II), 1 st Class (Part I).	2020–25 2019–20 2015–19
AWARDS & FUNDING	Canadian Neuroanalytic Scholarship Postdoctoral fellowship valued at CAD 140,000. UCL Early Career Neuroscience Prize Junior category, £500. Herchel Smith Scholarship \$80,000 fully funded scholarship to Harvard. Emmanuel College Cambridge John Mainhood Prize for high academic achievement.	2026–28 2023 2019 2016–18
SELECTED PUBLICATIONS	T. M. George* , M. Nakano*, K. Stachenfeld, C. Barry, & C. Clopath (2026) Repeated decoding is a fast, scalable method for neural latent and tuning curve discovery. Under review at Nature . Project . T. M. George (2025) SIMPL: Scalable and hassle-free optimisation of neural representations from behaviour. <i>International Conference on Learning Representations (ICLR)</i> . Paper Code . T. M. George (2023) A generative model of the hippocampal formation trained with theta driven local learning rules. <i>Advances in Neural Information Processing Systems (NeurIPS)</i> . Paper . E. J. Thompson, ..., T. M. George et al. (2026) Replay of procedural memory is independent of the hippocampus. Nature Neuroscience . Paper . T. M. George (2023) Theta sequences as eligibility traces: A biological solution to credit assignment. Tiny Papers Track at ICLR 2023 , Kigali, Rwanda (accepted for oral presentation). Paper . T. M. George , M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, & C. Barry (2024) RatInABox: An open source toolkit for modelling locomotion and neuronal activity in continuous environments. eLife . Paper Code . T. M. George* , W. de Cothi*, K. Stachenfeld, & C. Barry (2023) Rapid learning of predictive maps with STDP and theta phase precession. eLife ; 70 citations. Paper Code . T. M. George , G. E. Manucharyan, & A. F. Thompson (2021) Deep learning to infer eddy heat fluxes from sea surface height patterns of mesoscale turbulence. Nature Communications ; 80 citations. Paper . T. M. George & P. Liò (2019) Unsupervised Machine Learning for Data Encoding applied to Ovarian Cancer Transcriptomes. bioRxiv . Paper .	
SOFTWARE	RatInABox Creator and lead maintainer of an open-source scientific Python package that generates motion and neural data in spatial environments; 75,000+ PyPI downloads and an <i>eLife</i> Tools and Resources article. Code . SIMPL Originated and implemented a scalable neural latent-variable method, published at ICLR 2025 with open-source code and data. Code .	2022 2024
INDUSTRY	X AI Residency, Google Completed a five-month AI residency developing foundation models for geophysics at X, Google's moonshot factory, in Mountain View. ML consultant, Mltpl Developed machine-learning tools for automated business valuations.	2025 2023–24
TECHNICAL SKILLS	Programming Python, JAX, PyTorch, TensorFlow, bash and Julia. GitHub . Machine Learning Foundation models, representation learning, latent-variable models, RL, NLP, CNNs and RNNs. Languages English (Native), Spanish (DELE B1 equivalent).	
TEACHING & OUTREACH	TReND-CaMinA Co-founder and organizer of a recurring, fully funded computational neuroscience and ML school across Ghana, Rwanda, Zambia and Kenya. Co-developed its open-source curriculum and secured \$300,000 in funding. NeuroAI summer school University of Amsterdam, invited lecturer on model-free RL. Open-source teaching materials Created open teaching materials for deep learning, modern Hopfield networks, reinforcement learning, spatial representations, path integration and successor features. Materials .	2023–26 2024
SELECTED INVITED TALKS	Mila, Quebec AI Institute SIMPL; fast latent variable analysis for neural data. Oxford University, Cortex Club RatInABox (see open source).	2025 2024

	Google DeepMind, London Optimizing internal representations (NeuroLab workshop).	
	Imperial, Pint of Science Public outreach talk on "The language of the brain".	
	ICLR TinyPapers workshop oral Neural oscillations and eligibility traces, Kigali, Rwanda.	2023
	Google DeepMind, London The Helmholtz hippocampus (NeuroLab workshop).	
	UCL Neuroscience Symposium (prize winner) How hippocampus learns predictive maps?	
	Spring Hippocampal Research Conference, Verona RatInABox (see open source).	
	Mila Neural-AI reading group Biological Reinforcement Learning in the Hippocampus.	2022
ACADEMIC SERVICE	Area chair, ICLR 2023 and 2024 TinyPapers workshop.	
	Reviewer, NeurIPS 2023 Associative Memory & Hopfield Networks workshop.	
SELECTED POSTERS	COSYNE 2025 poster T. M. George , P. Glaser, K. Stachenfeld, C. Barry & C. Clopath. Repeated decoding optimises neural representations and is a SIMPL approach to finding latent variables in spiking data. Poster .	2025
	COSYNE 2024 poster C. Barry, M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, T. M. George (2024). RatInABox: A unified Python framework for modelling spatial behaviour and neural data.	
	COSYNE 2024 poster T. M. George , C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024). The Helmholtz Hippocampus: A biologically plausible generative model of the hippocampal formation.	
	NeurIPS 2023 poster T. M. George , C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024). The Helmholtz Hippocampus: A biologically plausible generative model of the hippocampal formation.	
	COSYNE 2024 poster T. M. George , W. de Cothi, K. Stachenfeld, C. Barry (2022). Rapid learning of predictive maps with STDP and theta phase precession.	
	SFN 2022 poster & Cosyne 2023 talk E. Thompson, L. Rollik, T. M. George* et al. Replay of motor sequences in DLS during consolidation using an unsupervised point process model.	
	SFN 2021 poster A. Onih, T. M. George , S. Nierwetberg & A. Akrami. Pupil dilation as a proxy for statistical learning in freely moving mice and humans.	
EARLIER PLACEMENTS	Caltech Summer Undergraduate Research Fellow with Prof. Andrew Thompson; the project led to a <i>Nature Communications</i> paper on deep-learning prediction of eddy heat fluxes.	2018
	Okinawa Computational Neuroscience Course (OCNC) Summer school participant.	2022
	Okinawa Institute of Science and Technology (OIST) Research intern with Prof. Tomoki Fukai.	2022